

계산이론(이,공) (CSE205-01)

학기	2026년 2학기
담당교수	전민석
이수구분	교선평수
학점	3.0
강의시간	화13:00-14:30(E7 - 241), 목13:00-14:30(E7 - 241)
강의실	
소속	기초학부

교과개요

This course provides a systematic introduction to the mathematical foundations of computer science. Building on mathematical preliminaries such as set theory, logic, and induction, the course covers three core topics. First, students study finite automata, regular expressions, and the properties of regular languages to understand the simplest computational models. Second, context-free languages and pushdown automata are explored to describe the grammatical structure of programming languages. Third, students examine the origins of computer science and study Turing machines — the theoretical model underlying modern computers — in both standard and extended forms, gaining insight into the nature of computability, undecidability (including the Halting Problem), and computational complexity through P, NP, and NP-completeness theory.

교과목표

Upon completion of this course, students will be able to:

1. Understand and explain the mathematical models of computation: finite automata, pushdown automata, and Turing machines.
2. Analyze the properties of regular and context-free languages, and apply the Pumping Lemma to classify formal languages.
3. Understand computability and undecidability through the lens of Turing machines, and prove representative undecidable problems such as the Halting Problem.
4. Understand P, NP, and NP-completeness, and classify problems from the perspective of computational complexity.
5. Develop rigorous mathematical reasoning and proof-writing skills applicable across all areas of computer science.

주차별 강의계획

- 1주차: Introduction
- 2주차: Mathematical Preliminaries
- 3주차: Finite Automata
- 4주차: Regular Languages & Regular Expressions
- 5주차: Properties of Regular Languages
- 6주차: Context-free Languages
- 7주차: Simplifications and Normal Forms
- 8주차: Midterm Exam
- 9주차: Pushdown Automata
- 10주차: Properties of Context-free Languages
- 11주차: The Origin of Computer Science

12주차:Standard Turing Machines

13주차:Extensions of Turing Machines

14주차:Undecidability / Limits of Algorithmic Computation

15주차:P, NP, and NP-completeness

16주차:Final Exam

수업방법

The instructor lectures with self-contained slides.

학습윤리

Students are expected to maintain the highest standards of academic honesty and integrity throughout this course. Violations of academic integrity, including plagiarism, cheating on exams will result in serious consequences ranging from failure on the assignment to failure in the course

교과정책(방침)

Attendance: Regular attendance is strongly encouraged and essential for success in this course. Students are responsible for all material covered in class, including announcements and assignment details.

Classroom Conduct: Students are expected to maintain a respectful and professional learning environment.

Disruptive behavior, including inappropriate use of electronic devices, will not be tolerated.